Financing a Concentrated Investment Boom: Evidence from AI

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Abstract

Evidence on the external financing of artificial-intelligence infrastructure describes the project margin, which need not characterise aggregate investment when incremental spending is concentrated among cash-rich firms. For the 500 largest U.S. public companies, internal cash coverage of cumulative incremental capital expenditure over 2022–2025 peaks at the five largest investors and declines as the sample widens, from 118% to −7% for the remaining 217. The top ten account for 52.5% of the increase at 104% coverage. A pre-specified 10-K measure of AI exposure yields the same conclusion. We report an accounting decomposition of realised flows, not a causal effect.

Keywords: investment concentration, internal finance, artificial intelligence, corporate investment, aggregation
JEL: G31, G32, E22, O32

1. Introduction

The buildout of artificial-intelligence infrastructure has prompted concern about leverage. Recent work documents its financing through private credit, infrastructure funds, off-balance-sheet vehicles and hardware-collateralised lending, and asks what the exposures imply for financial stability (Van Nieuwerburgh, 2026; Thelisson, 2026; Ledesma Padilla, 2026).

Prior work shows that corporate investment is highly concentrated and that cash flow is especially important for the largest investors (Grullon et al., 2018; Gutierrez and Philippon, 2017; Autor et al., 2020; Peters and Taylor, 2017), and that internal funds and investment covary at the firm level, though the interpretation of that covariation is debated (Fazzari et al., 1988; Kaplan and Zingales, 1997; Almeida et al., 2004; Myers and Majluf, 1984). We connect these two literatures by asking a different question: which firms account for the realised increase in investment during the buildout, and how does their incremental internal cash generation compare with that incremental investment? The aggregate answer differs sharply from the financing pattern of the typical firm, because the increase is concentrated among firms whose operating cash flow rose at least as rapidly.

The setting here is extreme: among the 500 largest U.S. public companies the Herfindahl index of capital expenditure rises from 173 in 2021 to 309 in 2025, while excluding the five largest spenders it moves only from 95 to 104. Those five—Alphabet, Amazon, Oracle, Meta

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and Microsoft, selected mechanically as those with the largest absolute increase in real capital expenditure—account for 67.3% of the endpoint increase in annual capital expenditure between 2021 and 2025, and 38.7% of cumulative incremental capital expenditure over 2022–2025.

We find that internal cash coverage is highest at the top of the distribution of incremental investment and declines steadily as the sample expands beyond the five largest investors. Cumulating incremental flows over 2022–2025, the ratio of incremental operating cash flow to incremental capital expenditure runs from 118% for the top five to −7% for the remaining 217 firms with rising investment (Table 1). Net debt rises alongside investment down the distribution: those 217 firms took on $280 billion of cumulative net new debt, against −$3 billion for the top five.

Our contribution is an aggregation result. Financing at the project margin need not characterise aggregate investment when incremental spending is concentrated among firms whose internal cash generation rises at least as fast as investment. The top ten firms account for 52.5% of cumulative incremental capital expenditure and have coverage of 104%. The distinction from Grullon et al. (2018) is one of object: that work concerns the determinants of investment and its long-run concentration, estimating how cash flow predicts investment, whereas ours is an ex-post coverage ratio for one boom, establishing financing capacity rather than identifying cash flow as a cause. The evidence therefore relocates rather than contradicts the leverage concern: it belongs with smaller participants and project-finance structures rather than with the aggregate. This is an accounting decomposition of realised flows. We cannot trace individual dollars, and we do not identify what any firm would otherwise have done.

2. Data and methods

Firm-level data come from the SEC's XBRL *frames* interface for 2013–2025. Filers do not use a single tag for a given concept, and relying on one is unsafe: Amazon, the largest U.S. capital spender, reports under `PaymentsToAcquireProductiveAssets` rather than `PaymentsToAcquirePropertyPlantAndEquipment`, as do 448 other firms accounting for $404 billion of 2025 capital expenditure. We take the maximum across alternative tags for every series. Measuring the buildout by total capital expenditure rather than an AI-specific subaccount follows established practice (Board of Governors, 2026; Van Nieuwerburgh, 2026).

The universe is the 500 U.S. registrants with the largest fiscal-2024 revenue, combined revenue $19.75 trillion—the Fortune 500's ranking basis, excluding non-filers. Listed firms exclude a growing private segment (Doidge et al., 2017). Figures are deflated to 2025 dollars using the BLS producer price index for private capital equipment (WPUFD4131).

Two measurement points require statement. The SEC assigns each fiscal year to the calendar year containing most of its months, so for off-calendar filers a year label is not a calendar year: Microsoft's observations are June-ending years and Oracle's May-ending, so Oracle's 2025 observation covers June 2025 to May 2026. We retain the SEC convention, which shifts Oracle's window forward relative to the other four. Second, every constructed observation for the five firms reconciles to a 10-K value for the same fiscal period: 48 of 49 match the originally filed figure and the 49th matches Meta's restated 2021 capital expenditure, which is the value the interface returns (Appendix B).

The exercise cumulates incremental flows rather than comparing two annual observations. For each series X and firm i,

$$\Delta X_i \;=\; \sum_{t=2022}^{2025} \big(X_{it} - \bar{X}_i\big), \tag{1}$$

$$\text{Coverage}_i = \frac{\Delta\text{OCF}_i}{\Delta\text{CAPEX}_i}, \tag{2}$$

where $\bar{X}_i$ is either the 2021 value (baseline A) or the 2017–2021 average (baseline B). The ratio requires an observed baseline as well as a positive denominator: a firm absent in 2021 would otherwise be assigned a baseline of zero, making its entire subsequent spending appear incremental. Twenty-five firms fail that condition and are excluded, removing \$36 billion of spurious increment. The condition holds for 267 firms; firms whose investment fell are excluded rather than assigned a signed ratio. Because R&D is an operating expense already deducted in arriving at operating cash flow, it cannot enter as a separately financed use and is reported as a memorandum item.

The five focal firms are selected mechanically from financial statements, not from AI disclosures. Contemporaneous company filings independently identify AI-related compute capacity and data-centre infrastructure as drivers of the investment acceleration for three of the five; Appendix A records the evidence firm by firm, including the two cases where the filings do not support that reading. Coverage above 100% establishes that internally generated cash grew by more than investment did, not that particular dollars financed particular assets. Every reported number is produced from frozen, hashed source files by a single script, and the manuscript is gated against the resulting claims file.

3. Results

3.1. Coverage declines down the investment distribution

Table 1 reports the central result. Ranking the 267 firms with rising investment by their cumulative incremental capital expenditure, internal cash coverage peaks at the five largest investors and then falls at every successive cut, from 118% for the top five to −7% for the remaining 217, while cumulative net new debt rises from −\$3 billion to \$280 billion. The top ten firms account for 52.5% of the aggregate increase at a coverage ratio of 104%.

Coverage is not monotone across the first three cuts: 67% for the largest single investor and 70% for the top three, before rising to 118% at the top five. This reflects the firm-level dispersion in Section 3.2: Alphabet, Meta and Oracle lie between 63% and 79%, while Microsoft and Amazon, ranked fourth and fifth, are at 122% and 344%. The decline is a property of expanding the sample beyond the five largest investors, and that is the claim we make.

The gradient is the mechanism. Aggregate financing structure is a capital-expenditure-weighted average of firm-level financing structures, so an aggregate dominated by a few internally financed firms will look internally financed even when most firms in the distribution are borrowing. Firms with coverage below 50% account for 30.4% of positive incremental capital expenditure, but not for where the aggregate increase occurred.

The pattern is not a mechanical consequence of the five focal firms. Excluding them, 34.0% of positive incremental capital expenditure still occurred at firms with coverage at or above 100%, against 33.3% including them. What the five contribute is weight at the top of the distribution, not a different relationship between rank and coverage.

3.2. The aggregate for the top of the distribution

For the five largest incremental investors, cumulative incremental operating cash flow over 2022–2025 was \$438 billion against \$371 billion of cumulative incremental capital expenditure, a coverage ratio of 118%, with cumulative net new debt of −\$3 billion and net new equity of

$2 billion. On the 2017–2021 average baseline the corresponding figures are $762 billion, $588 billion, coverage 130%, and net new debt $37 billion. Both baselines give the same answer.

Coverage exceeds 80% omitting any single firm, ranging from 81% (omitting Amazon) to 136% (omitting Alphabet), and exceeds 100% in four of the five cases. Firm-level coverage is nonetheless dispersed—63% at Oracle, 67% at Alphabet, 79% at Meta, 122% at Microsoft and 344% at Amazon—so sufficiency is a property of the group aggregate and not of every constituent. Appendix B reports the firm-level decomposition and the leave-one-out calculation.

3.3. Attribution: AI-exposed firms identified independently

The gradient above is a fact about corporate investment rather than about AI: the firms at the bottom of the distribution are utilities, airlines, pipelines and energy producers, whose borrowing is unrelated to compute. Attribution therefore requires identifying AI-exposed firms without reference to any financing outcome. We score each registrant's most recent 10-K against a dictionary of AI terminology fixed before any subsample result was computed, following established practice in building firm-level measures from filing text (Loughran and McDonald, 2011; Hoberg and Phillips, 2016; Ante and Saggu, 2025). AI intensity is dictionary occurrences per 10,000 words; details are in Appendix D. The measure identifies the expected firms without conditioning on industry: the highest-scoring names are NVIDIA, Amazon, Meta, Advanced Micro Devices, Microsoft, Oracle and Alphabet.

Within the top decile of AI intensity (27 firms, $401 billion of cumulative incremental capital expenditure) coverage is 128% and cumulative net new debt is −$24 billion; at the top-quintile cutoff (53 firms) coverage is 138% (Table 1, panel B). Coverage exceeds 100% at every cut and net new debt is zero or negative throughout, against −7% and $280 billion across the full distribution, so observed net debt accumulation is concentrated outside the AI-exposed group. The five largest investors account for 92.5% of the top decile's incremental capital expenditure, so this corroborates the main result on an independently defined universe rather than testing it independently; excluding them, coverage among the remaining 22 firms is 248%.

3.4. R&D and payout

The investment increase did not coincide with reduced research: cumulative incremental R&D at the five firms was $142 billion on baseline A and $229 billion on baseline B, and among firms reporting research at both endpoints capital-expenditure growth is positively associated with R&D growth. Payout results are sensitive to the choice of baseline and we draw no inference from them. Both sets of estimates are reported in Appendix F.

3.5. The margin narrowed

The result is bounded in time. Capital expenditure absorbed 31% of the five firms' operating cash flow in 2017, 46% in 2021 and 70% in 2025, so internal financing slack was substantially lower at the end of the sample than at its start. That is consistent with the shift toward external financing the recent literature documents, and locates it after our window rather than within it.

4. Conclusion

The financing of an investment boom cannot be inferred from the financing of its marginal projects. The artificial-intelligence buildout illustrates why. Internal cash coverage of incremental investment peaks at the five largest incremental investors and declines at every successive

Table 1: Internal cash coverage of incremental investment, cumulative flows 2022–2025

	Firms	Share of increase (%)	Coverage (%)	Net new debt ($bn)
Panel A. By rank in incremental capital expenditure				
Top 1	1	10.1	67	0
Top 3	3	26.2	70	60
Top 5	5	38.7	118	−3
Top 10	10	52.5	104	29
Top 20	20	63.9	84	−69
Top 50	50	81.5	55	148
Remaining firms	217	18.5	−7	280
Panel B. By AI intensity, measured from 10-K text				
Top decile	27	41.8	128	−24
excluding top five	22	3.1	248	−20
Top quintile	53	43.4	138	0
excluding top five	48	4.8	301	3

Notes. Sample: the 267 firms among the 500 largest U.S. registrants by fiscal-2024 revenue that report capital expenditure in both 2021 and 2025 and whose cumulative incremental capital expenditure over 2022–2025 is positive; their aggregate incremental capital expenditure is $959 billion. Increments follow equation (1) and coverage equation (2). Shares are of that $959 billion. Amounts in billions of 2025 dollars, deflated by BLS series WPUFD4131. AI intensity is dictionary occurrences per 10,000 words of the firm's most recent 10-K, with dictionary and cutoffs fixed before any financing result was computed (Appendix D). On the 2017–2021 average baseline the top five show coverage of 130% and net new debt of $37 billion.

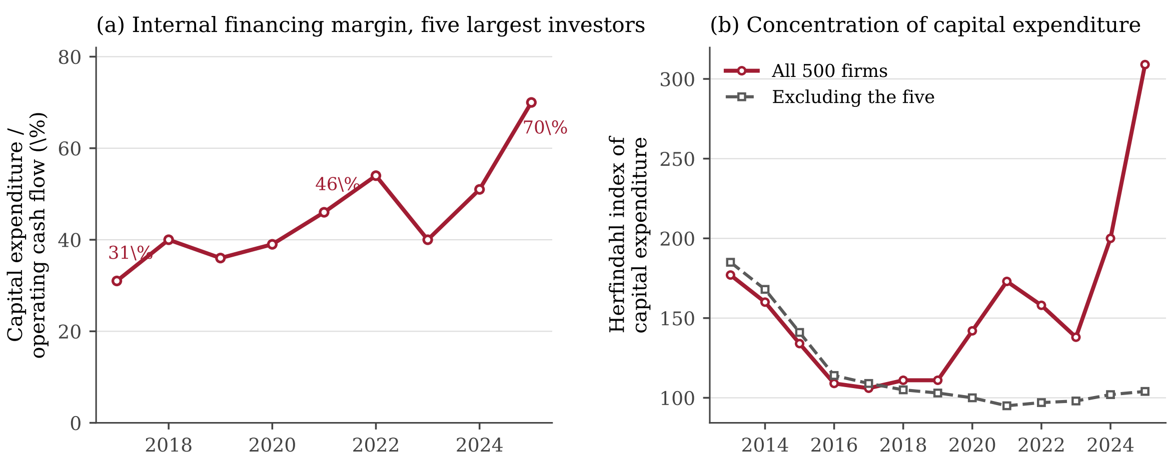

Figure 1: Left: capital expenditure as a share of operating cash flow, the five largest incremental investors, 2017–2025. Right: Herfindahl index of capital expenditure among the 500 largest U.S. public companies by revenue, with and without those five firms.

cut thereafter, from 118% at the top five to −7% among the remaining 217, while net new debt rises correspondingly. Because half the aggregate increase occurred at ten firms whose incremental cash generation exceeded their incremental investment, the aggregate is internally covered even though net debt rose alongside investment for most firms in the distribution.

This reconciles two literatures rather than adjudicating between them. Documented growth in private credit, structured vehicles and hardware-backed lending concerns a part of the distribution accounting for a minority of incremental corporate investment, and our result does not contest it; it locates the leverage concern with smaller participants and project-finance structures rather than with the aggregate. The qualification is temporal: capital expenditure absorbed 70% of the five firms' operating cash flow in 2025 against 31% in 2017, so the result need not persist.

Two limitations bound the reading. The exercise is an accounting decomposition of realised flows: it establishes that incremental internal cash generation exceeded incremental investment, not that particular dollars financed particular assets, and it does not identify counterfactual firm behaviour. And the universe is large public filers, so it is silent on private firms and on the special-purpose vehicles that much of the external-financing evidence concerns.

CRediT authorship contribution statement

Samir Chincholikar: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Robin Chawla:** Conceptualization, Validation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All code and derived data required to reproduce every table, figure and reported number are openly available in the replication archive. The underlying raw data are public: the U.S. SEC XBRL *frames* interface and Bureau of Labor Statistics series WPUFD4131.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) for code scaffolding, code review, minor grammatical edits and readability improvements. The tool was not used to generate, analyse or alter any datum, estimate, table or figure: all results are produced by the deposited scripts from the frozen source files. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. AI attribution: primary-source validation

The five firms are selected mechanically from financial statements, not from AI disclosures. This table records what their own filings say; quotations are verbatim.

Firm	Filing	Disclosure	Capex–AI link
Alphabet	10-K FY2025	"We continue to invest in capital expenditures as we scale our technical infrastructure, in particular for AI"; "we expect to significantly increase, relative to 2025, our investment in our technical infrastructure, including servers and network equipment, and data centers."	Explicit
Meta	10-K FY2025	"$69 billion of purchases of property and equipment as we continued to invest in servers, data centers, and network infrastructure"; "We anticipate making capital expenditures of approximately $115 billion to $135 billion in 2026 to support our AI efforts and core business."	Explicit
Microsoft	10-K FY2026	"We will continue to invest in capital expenditures to support growth in our cloud offerings and our investments in AI training and other infrastructure."	Explicit
Oracle	10-K FY2026	Capital expenditure appears in risk factors as "multi-year commitments for excess data center space and related capital expenditures" and retrofit needs from "technological changes in data center design and equipment."	Partial
Amazon	10-K FY2025	No sentence in the filing links capital-expenditure growth to AI or compute capacity; AI is discussed in competition and technology contexts only.	None

Sources. Filings are retrievable from SEC EDGAR at `https://www.sec.gov/Archives/edgar/data/` *CIK* / *accession* /, with: Alphabet, CIK 1652044, accession 0001652044-26-000018; Meta, 1326801, 0001628280-26-003942; Microsoft, 789019, 0001193125-26-323660; Oracle, 1341439, 0001193125-26-277521; Amazon, 1018724, 0001018724-26-000004.

Assessment. Three of five firms explicitly attribute capital-expenditure growth to AI or AI-supporting infrastructure in their 10-K. Oracle's capital-expenditure discussion is data-centre-specific but not AI-explicit. Amazon's 10-K contains no such statement, so its inclusion rests solely on the mechanical rule. The main text therefore claims only that contemporaneous disclosures independently identify AI-related compute and data-centre infrastructure as important drivers for most of the group, not that all incremental capital expenditure at these firms is AI expenditure.

Appendix B. Reconciliation to as-filed 10-K values

Each constructed observation for the five focal firms is matched to a 10-K-reported value for the *same fiscal period.* Matching on period rather than on year label is necessary because the SEC assigns a fiscal year to the calendar year containing most of its months.

	Count	
Observations reconciled (5 firms, 7 series, 2 years, where reported)	49	
Exact match to originally filed value	48	98%
Matches restated value	1	Meta capital expenditure 2021
Unexplained discrepancies	0	

The single non-identical case is not an extraction error. Meta reported 2021 capital expenditure of $18.567bn in its FY2021 10-K (filed 2022-02-03) and $18.690bn for the same period in its FY2022 and FY2023 10-Ks (accession 0001326801-23-000013 and 0001326801-24-000012). The SEC interface returns the most recently reported figure, and our series uses it. No constructed value differs from the corresponding filed value once restatement is taken into account.

Fiscal-period alignment. Microsoft's observations are fiscal years ending in June and Oracle's fiscal years ending in May. Oracle's 2021 observation therefore covers 2021-06-01 to 2022-05-31 and its 2025 observation covers 2025-06-01 to 2026-05-31; the latter extends five months beyond the calendar-year endpoint used for Alphabet, Amazon and Meta. We retain the SEC convention. Because Oracle's investment rose steeply in that fiscal year, this shifts its contribution later rather than earlier, and the leave-one-out calculation in Appendix C shows group coverage of 131% with Oracle omitted, against 118% including it.

Appendix C. Firm-level decomposition and leave-one-out

Cumulative incremental flows, 2022–2025, baseline A (2021), billions of 2025 dollars.

Firm	ΔCapex	ΔOCF	Net debt	Net equity	ΔR&D	ΔRepurch.	Coverage
Alphabet	97	65	0	0	53	0	67%
Meta	82	65	60	0	64	−103	79%
Oracle	73	46	0	2	5	−74	63%
Microsoft	68	83	0	0	20	−35	122%
Amazon	52	179	−64	0	0	7	344%
Leave-one-out: group coverage with the named firm omitted							
Omit Alphabet	274	373					136%
Omit Meta	289	372					129%
Omit Oracle	298	391					131%
Omit Microsoft	303	355					117%
Omit Amazon	319	259					81%

Amazon reports no separate R&D line, so its ΔR&D is zero by construction rather than by measurement.

Appendix D. AI-exposure measure: pre-specification and construction

Source. The most recent 10-K for each sample firm, retrieved from SEC EDGAR via the submissions API. HTML is stripped to text and lower-cased; 261 of the 267 sample firms yielded a scoreable filing (six had no retrievable 10-K primary document).

Pre-specification. The dictionary and cutoffs below were fixed in the collection script before any subsample financing result was computed, and the script header records this. No term was added, removed or reweighted after inspecting a result.

Dictionary. Model and method vocabulary: *artificial intelligence, machine learning, generative ai, deep learning, neural network, large language model, foundation model, ai model, ai capabilities, ai infrastructure, accelerated computing.* A separate infrastructure list (*data center/centre, graphics processing unit, gpu, compute capacity, accelerator, training cluster, inference capacity*) is recorded but not used in the primary cut.

Score. AI intensity$_i$ = 10,000 × (dictionary occurrences)/(total words). Distribution across the sample: median 0.64, 80th percentile 1.35, 90th percentile 2.21.

Cutoffs. AI-exposed = top decile (threshold 2.21, $n = 27$); robustness at top quintile (threshold 1.35, $n = 53$).

Face validity. The ten highest-scoring firms are NVIDIA, Amazon, Meta, Advanced Micro Devices, Microsoft, Medline, Oracle, Alphabet, Target and Workday. Seven of the ten are compute or platform firms. The measure also admits firms that discuss AI heavily without building infrastructure—Medline, Target, Workday—which is a known cost of a term-frequency filter and is not corrected ex post. Because these firms are small incremental investors, they carry little weight: the five largest incremental investors account for 92.5% of the top decile's incremental capital expenditure.

Appendix E. Alternative group definitions

Group	ΔCapex	ΔOCF	Net debt	Coverage	Share of increase
Top 3	251	176	60	70%	26.2%
Top 5	371	438	−3	118%	38.7%
Top 10	503	522	29	104%	52.5%
Top 20	613	517	−69	84%	63.9%

Groups are ranked by cumulative incremental capital expenditure within the 267-firm analysis sample, on the same basis as Table 1. Amounts in billions of 2025 dollars; shares are of the $959 billion aggregate cumulative increment.

Appendix F. R&D and payout estimates

Among the 153 firms reporting positive R&D, capital expenditure and revenue at both endpoints—165 of the 500 report R&D in both years, so this is not a statement about the full universe—the slope of real R&D growth on real capital-expenditure growth over 2021–2025 is 0.484 ($t = 11.4$), and 0.111 net of real revenue growth ($t = 2.7$). Excluding the five firms it is 0.520 ($t = 11.8$), and the same slope is 0.517 over 2017–2021 and 0.501 over 2013–2017, so the association predates the boom.

Two payout narratives are baseline-sensitive. Share repurchases at the five firms fall $206 billion cumulatively on baseline A but rise $78 billion on baseline B, because 2021 was a peak repurchase year. A contrast in which the other 495 firms appear to borrow while their cash flow falls reverses sign entirely across baselines (−81% against 411%). We draw no conclusions from either.

Appendix G. Data construction

Capital expenditure unions `PaymentsToAcquirePropertyPlantAndEquipment` and `PaymentsToAcquireProductiveAssets`; operating cash flow unions `NetCashProvidedByUsedInOperatingActivities` with its continuing-operations variant; repurchases and dividends likewise union their alternates. Single-tag construction omits Amazon and 448 other filers representing $404 billion of 2025 capital expenditure. Deflator: BLS PPI WPUFD4131. All source files are frozen and SHA-256 hashed; every number in the manuscript is verified against a machine-generated claims file by an automated build gate.